# Project #4: Engineering Timesheet

CIVE 202 – Civil Engineering Analysis II

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Project Duration: April 2 – April 21, 2026

| Activity | Awad (hrs) | Johnson (hrs) | Wiggins (hrs) | Total (hrs) |
|---|---|---|---|---|
| **Task 1: Data Loading & Verification** | | | | |
| Load NRI Census Tract and County data | 2.0 | 2.0 | 0.0 | **4.0** |
| Load SVI datasets (CA and CO) | 0.0 | 0.0 | 2.0 | **2.0** |
| Verify row counts, column counts, data types | 1.5 | 0.0 | 0.0 | **1.5** |
| Review NRI Data Dictionary | 0.0 | 1.0 | 1.0 | **2.0** |
| **Task 2: Data Cleaning & Integration** | | | | |
| Handle -999 sentinels in RPL_THEMES | 2.0 | 0.0 | 0.0 | **2.0** |
| Standardize FIPS codes for merge | 0.0 | 1.5 | 0.0 | **1.5** |
| Merge NRI and SVI datasets | 2.0 | 1.0 | 0.0 | **3.0** |
| Verify merge completeness | 0.0 | 0.0 | 1.0 | **1.0** |
| **Task 3: Alternative Risk Score Calculation** | | | | |
| Implement alternative risk formula | 3.0 | 0.0 | 0.0 | **3.0** |
| Calculate scores for all valid tracts | 1.5 | 0.0 | 0.0 | **1.5** |
| Bin scores into qualitative categories | 0.0 | 1.5 | 0.0 | **1.5** |
| Sensitivity analysis (weight variation) | 1.5 | 1.5 | 0.0 | **3.0** |

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|---|---|---|---|---|
| **Task 4: Summary Tables & Figures** | | | | |
| Cross-tabulation tables | 0.0 | 2.0 | 0.0 | **2.0** |
| Distribution comparison plots | 0.0 | 0.0 | 2.0 | **2.0** |
| EAL vs. SVI scatter plot | 0.0 | 1.0 | 1.0 | **2.0** |
| **Task 5: GeoPandas Mapping** | | | | |
| Download Census Tract shapefile | 0.0 | 0.0 | 1.0 | **1.0** |
| Merge risk data with shapefile | 2.0 | 0.0 | 0.0 | **2.0** |
| Generate choropleth maps (4 maps) | 2.5 | 0.0 | 1.5 | **4.0** |
| **Task 6: Documentation** | | | | |
| Scope of Work | 3.0 | 0.0 | 0.0 | **3.0** |
| Gantt chart | 1.5 | 0.0 | 0.0 | **1.5** |
| GeoPandas Investigation | 1.0 | 0.0 | 0.0 | **1.0** |
| Risk Definition document | 1.5 | 0.0 | 0.0 | **1.5** |
| Annotated code document | 0.0 | 3.0 | 0.0 | **3.0** |
| GitHub repository and README | 1.5 | 0.0 | 0.0 | **1.5** |
| **Task 7: Technical Report** | | | | |
| Methodology section | 2.0 | 1.5 | 0.0 | **3.5** |
| Results and Discussion | 0.0 | 2.0 | 2.0 | **4.0** |
| Ethical implications and bias analysis | 1.5 | 1.0 | 1.5 | **4.0** |
| References and final formatting | 1.0 | 0.0 | 0.0 | **1.0** |
| **Final Review & Submission** | | | | |
| Final review | 1.0 | 0.5 | 0.5 | **2.0** |
| **TOTAL HOURS** | **32.0** | **19.5** | **13.5** | **65.0** |

**Certification:**

We certify that the hours reported above are an accurate representation of the time spent on each activity for Project #4.